Coworks-BSL

EV Speed Profile Optimization with Battery Degradation and Thermal Management

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Overall Architecture: main.m

1. Traffic rollout

- `VissimInitconfig`, `VissimSetupScenario`
- `VissimRunInit`: warm-up and target vehicle detection
- `VissimRollout`: $N_p + 1$-step prediction window from VISSIM

2. Optimization and model rollout

- `get_bounds`: gap-based speed and acceleration bounds
- `get_MPCsolver`: constrained speed-profile MPC
- `get_Optspeeds`: baseline/optimized EV rollout over the window

3. Replay and logging

- `VissimRewind`: return to the current control step
- `VissimReplayOpt`: apply the optimized command
- `get_HistoryInit`, `get_HistoryLog`, `get_plot`

VISSIM–MATLAB Block Diagram

VISSIM

- `VissimSetupScenario`
- `VissimRunInit`
- `VissimRollout`
- ego speed, position, gap, lane states

→

MATLAB

- `get_bounds`
- `get_MPCsolver`
- `get_Optspeeds`
- `get_evmodel`
- speed optimization with power, thermal, and aging models

→

Replay / Logging

- `VissimRewind`
- `VissimReplayOpt`
- `get_HistoryLog`
- applied trajectory and battery-state update

Prediction window from VISSIM → optimization in MATLAB → replay and history update

Models Considered In The Codebase

1. Traffic interaction model

- ego speed, position, acceleration, front gap, rear gap
- prediction window generated from VISSIM rollout
- gap-based bounds for feasible speed planning

2. Powertrain model

- road slope interpolation from `StudyArea_SlopeMap.csv`
- longitudinal torque balance with drag, grade, and rolling resistance
- traction and regenerative power maps

3. Battery model

- pack/module/cell hierarchy
- OCV and resistance model
- thermal management and current solve
- aging and normalized degradation cost

State Flow Across The Closed Loop

Traffic states

- `Pred_speds`: ego motion, gaps, and neighbor speeds
- used by `get_bounds` and `get_MPCsolver`

Battery states

- `batt`: $Q_{\mathrm{bat}}, T, Ah_{\mathrm{ir}}, Ah_{\mathrm{th}}, Ah_{\mathrm{eff}}$
- prediction-side copies in `get_Optspeeds`
- real state update in `VissimReplayOpt`

Logged outputs

- prediction window: baseline vs optimized
- replay trajectory: applied vs baseline
- power, current, SOC, temperature, and aging indicators

Road Slope And Position: get_slope.m

Absolute longitudinal position

$$s(t) = d(t) + x_0 \tag{1}$$

- $s(t)$: absolute longitudinal distance along the route
- $d(t)$: cumulative traveled distance over the rollout
- x_0: initial position

Interpolated road slope

$$g(s(t_{\text{init}})) = \text{interp1}(\text{degmap}(:,1), \text{degmap}(:,2), s(t_{\text{init}}))\,, \tag{2}$$

with slope clamping

$$g_{\text{slope}} = \max(\min(g, 0.3)\,, -0.3) \tag{3}$$

The road angle is then calculated as

$$\theta_{\text{road}} = \tan^{-1}(g_{\text{slope}}) \tag{4}$$

Longitudinal Torque Terms: get_evmodel.m

Torque components

$$T_{\mathrm{acc}} = \frac{m_{\mathrm{eq}} a R_t}{\gamma} \tag{5}$$

$$T_{\mathrm{air}} = \frac{k_{\mathrm{air}} (v + v_w)^2 R_t}{\gamma} \tag{6}$$

$$T_{\mathrm{grad}} = \frac{mg \sin(\theta_{\mathrm{road}}) R_t}{\gamma} \tag{7}$$

$$T_{\mathrm{roll}} = \frac{\mathrm{RRC}\, mg \cos(\theta_{\mathrm{road}}) R_t}{\gamma} \tag{8}$$

total mechanical torque

$$T_{\mathrm{mech}} = T_{\mathrm{acc}} + T_{\mathrm{air}} + T_{\mathrm{grad}} + T_{\mathrm{roll}} \tag{9}$$

Motor Torque And Speed: usr_MOVE2MT.m

Map input torque

$$T_{\mathrm{mot}} = \frac{\left(ma + \frac{1}{2}(0.5)(1.188)(2.06)\left(\frac{\mathrm{SP}}{3.6}\right)^2 + \mathrm{RRC}mg\right)R}{\mathrm{gear}} \tag{10}$$

- SP: vehicle speed in km/h
- T_{mot}: torque input to the power map

Motor speed

$$\omega_m = \left|\frac{v}{R_t}\right|\gamma \tag{11}$$

$$\mathrm{rpm} = \omega_m \frac{30}{\pi} \tag{12}$$

Traction Power Map: usr_MT2BC_sys.m

Polynomial inputs

$$\boldsymbol{x} = \begin{bmatrix} 1 & MT & MT^2 & MT^3 \end{bmatrix} \tag{13}$$

$$\boldsymbol{y} = \begin{bmatrix} 1 & SP & SP^2 & 1 \end{bmatrix} \tag{14}$$

Map output

$$\text{consumption} = \sum \boldsymbol{y} \odot (\text{sys}\, \boldsymbol{x}^\top)^\top \tag{15}$$

traction electrical power

$$P_{\text{raw}} = 10^3 \cdot \text{usr_MT2BC_sys}(T_{\text{mot}}, v_{\text{km/h}}, \text{BPsys}) \tag{16}$$

Regenerative Braking: usr_REGENBP.m

positive traction

$$P_{\mathrm{trac}} = P_{\mathrm{raw}}, \qquad P_{\mathrm{raw}} \geq 0 \tag{17}$$

regenerative operation

$$\mathrm{coeff} = \exp\left(\frac{0.0411}{|a|}\right)^{-1} \tag{18}$$

$$P_{\mathrm{trac}} = 0.125 \cdot \mathrm{coeff} \cdot P_{\mathrm{raw}}, \qquad P_{\mathrm{raw}} < 0 \tag{19}$$

Battery Model And SOC: get_evmodel.m

Battery hierarchy

$$96\text{s}24\text{p} = \underbrace{(16\text{s}24\text{p})}_{1module} \times 6$$

- $N_p = 24$: parallel cells per module
- $N_{\text{s,mod}} = 16$: series cells per module
- $N_{\text{mod,s}} = 6$: series modules per pack

Capacity variables

$$Q_{\text{cell}} = \frac{Q_{\text{bat}}}{N_p} \tag{20}$$

$$Q_{\text{mod}} = Q_{\text{bat}} \tag{21}$$

State of charge

$$\text{SOC} = \min\left(\max\left(\frac{Q_{\text{bat}}}{Q_{\text{bat,nom}}}, 0\right), 1\right) \tag{22}$$

Resistance And OCV: get_evmodel.m

Module internal resistance

$$T_{\mathrm{mod},^\circ C} = T_{\mathrm{pack}} - 273.15 \tag{23}$$

$$Ah_{\mathrm{mod}} = \max(Ah_{\mathrm{ir}}, Ah_{\mathrm{floor}}) \tag{24}$$

$$R_{\mathrm{mod}} = \mathrm{feval}(\mathrm{Fit_IR}, T_{\mathrm{mod},^\circ C}, Ah_{\mathrm{mod}}) \tag{25}$$

$$R_{\mathrm{pack}} = R_{\mathrm{mod}}\, N_{\mathrm{mod,s}} \tag{26}$$

2. Module OCV polynomial

$$x = \frac{Q_{\mathrm{mod}} - \mu_1}{\mu_2} \tag{27}$$

$$\mathrm{OCV}_{\mathrm{mod}} = p_1 x^6 + p_2 x^5 + p_3 x^4 + p_4 x^3 + p_5 x^2 + p_6 x + p_7 \tag{28}$$

$$\mathrm{OCV}_{\mathrm{pack}} = \mathrm{OCV}_{\mathrm{mod}}\, N_{\mathrm{mod,s}} \tag{29}$$

Cooling Power Demand and Controller Design: get_evmodel.m

Cooling power command

$$e_T = T_{\text{pack}} - T_{\text{ref}} \tag{30}$$

$$P_{\text{cool}} = \begin{cases} P_0 + K_{p,T}(e_T - T_{\text{deadband}}), & e_T > T_{\text{deadband}} \\ 0, & \text{otherwise} \end{cases} \tag{31}$$

$$P_{\text{cool}} \in [0, P_{\text{cool,max}}] \tag{32}$$

- e_T: temperature error
- T_{ref}: reference temperature of the cooling system
- T_{deadband}: deadband for cooling activation
- P_0: baseline cooling power at activation threshold
- $K_{p,T}$: proportional gain for cooling power

Cooling Power Consumption Model: get_evmodel.m

Cooling heat removal [1]

$$Q_{\text{cool}} = \eta_{\text{cool}} P_{\text{cool}} \tag{33}$$

- η_{cool}: cooling system efficiency

Battery power demand

$$P_{\text{req}} = P_{\text{trac}} + P_{\text{cool}} \tag{34}$$

[1] Source text here

Battery Current Solve: get_evmodel.m

1. Pack current equation

$$P_{\text{req}} = \text{OCV}_{\text{pack}} I_{\text{batt}} - R_{\text{pack}} I_{\text{batt}}^2 \tag{35}$$

2. Discriminant

$$\Delta = \text{OCV}_{\text{pack}}^2 - 4R_{\text{pack}} P_{\text{req}} \tag{36}$$

3. Current solution

$$I_{\text{batt}} = \frac{\text{OCV}_{\text{pack}} - \sqrt{\max(\Delta, 0)}}{2R_{\text{pack}}} \tag{37}$$

$$I_{\text{cell}} = \frac{I_{\text{batt}}}{N_p} \tag{38}$$

Thermal And Charge States: get_evmodel.m

1. Thermal terms

$$Q_{\text{joule}} = I_{\text{batt}}^2 R_{\text{pack}} \tag{39}$$

$$Q_{\text{env}} = hA(T_{\text{pack}} - T_{\text{amb}}) \tag{40}$$

$$Q_{\text{cool}} = \eta_{\text{cool}} P_{\text{cool}} \tag{41}$$

2. Temperature update

$$T_{k+1} = T_k + \frac{\Delta t}{C_{\text{th}}} \left(Q_{\text{joule}} - Q_{\text{env}} - Q_{\text{cool}}\right) \tag{42}$$

Charge And Throughput Update: get_evmodel.m

3. Charge and throughput update

$$Q_{\mathrm{bat},k+1} = \min\left(\max\left(Q_{\mathrm{bat},k} - I_{\mathrm{batt}}\frac{\Delta t}{3600}, 0\right), Q_{\mathrm{bat,nom}}\right) \tag{43}$$

$$Ah_{\mathrm{ir},k+1} = Ah_{\mathrm{ir},k} + |I_{\mathrm{batt}}|\frac{\Delta t}{3600} \tag{44}$$

$$Ah_{\mathrm{th},k+1} = Ah_{\mathrm{th},k} + |I_{\mathrm{cell}}|\frac{\Delta t}{3600} \tag{45}$$

Aging Severity Model: get_evmodel.m

1. SOC-dependent coefficients

$$(\alpha_{\text{age}}, \beta_{\text{age}}) = \begin{cases} (1287.6, 6356.3), & \text{SOC}_{k+1} \leq 0.45 \\ (1385.5, 4193.2), & \text{SOC}_{k+1} > 0.45 \end{cases} \tag{46}$$

2. Cell C-rate

$$Q_{\text{cell,nom}} = \frac{Q_{\text{bat,nom}}}{N_p} \tag{47}$$

$$I_c = \frac{|I_{\text{cell}}|}{Q_{\text{cell,nom}}} \tag{48}$$

3. Aging factor

$$f_{\text{age}} = (\alpha_{\text{age}}\text{SOC}_{k+1} + \beta_{\text{age}}) \exp\left(\frac{-31700 + 163.3 I_c}{R_{\text{gas}} T_{k+1}}\right) \tag{49}$$

$$Q_{\text{loss}} = f_{\text{age}}\, Ah_{\text{th},k+1}^{n_{\text{age}}} \tag{50}$$

Normalized Aging Cost: get_evmodel.m

1. Reference aging factor

$$f_{\mathrm{age,ref}} = (\alpha_{\mathrm{ref}}\mathrm{SOC}_{\mathrm{ref}} + \beta_{\mathrm{ref}}) \exp\left(\frac{-31700 + 163.3C_{\mathrm{rate,ref}}}{R_{\mathrm{gas}}T_{\mathrm{ref,age}}}\right) \tag{51}$$

2. Lifetime scaling

$$\gamma = \left(\frac{Q_{\mathrm{ref,\%}}}{f_{\mathrm{age}}}\right)^{1/n_{\mathrm{age}}} \tag{52}$$

$$\Gamma = \left(\frac{Q_{\mathrm{ref,\%}}}{f_{\mathrm{age,ref}}}\right)^{1/n_{\mathrm{age}}} \tag{53}$$

$$\sigma = \frac{\Gamma}{\gamma} \tag{54}$$

Effective Aging Throughput: get_evmodel.m

3. Effective throughput and stage cost

$$Ah_{\mathrm{eff},k+1} = Ah_{\mathrm{eff},k} + \sigma |I_{\mathrm{cell}}| \frac{\Delta t}{3600} \tag{55}$$

$$\mathrm{AgeCost}_k = \sigma |I_{\mathrm{cell}}| \tag{56}$$

Prediction Variables: get_Optspeeds.m

1. Decision variable

$$\boldsymbol{x} = \begin{bmatrix} v_2 & v_3 & \cdots & v_{N_p+1} \end{bmatrix}^\top \tag{57}$$

2. Speed vector

$$\boldsymbol{v}_{\text{all}} = \begin{bmatrix} v_0 & \boldsymbol{x}^\top \end{bmatrix}^\top \tag{58}$$

3. Step acceleration

$$a_k = \frac{v_{k+1} - v_k}{\Delta t_k} \tag{59}$$

4. Distance propagation

$$d_k = d_{k-1} + v_k \Delta t_k \tag{60}$$

Objective Function: get_MPCsolver.m

1. Total objective

$$J = \sum_{k=1}^{N_p} \left[\alpha J_{\text{energy},k} + (1-\alpha) J_{\text{deg},k} \right] \tag{61}$$

2. Stage energy term

$$E_{\text{trac},k} = P_{\text{trac},k} \frac{\Delta t_k}{3600} \tag{62}$$

$$E_{\text{cool},k} = P_{\text{cool},k} \frac{\Delta t_k}{3600} \tag{63}$$

$$E_{\text{ref}} = V_{\text{nom}} Q_{\text{bat,nom}} \tag{64}$$

$$J_{\text{energy},k} = \frac{w_{\text{trac}} E_{\text{trac},k} + w_{\text{cool}} E_{\text{cool},k}}{E_{\text{ref}}} \tag{65}$$

Aging Cost In The Objective: get_MPCsolver.m

1. Instantaneous aging term from EV model

$$\mathrm{AgeCost}_k = \sigma_k |I_{\mathrm{cell},k}| \tag{66}$$

2. Current normalization

$$I_{\mathrm{cell,ref}} = \frac{P_{\max}}{V_{\mathrm{nom}} N_p} \tag{67}$$

3. Normalized aging cost

$$J_{\mathrm{deg},k} = \frac{\mathrm{AgeCost}_k}{I_{\mathrm{cell,ref}}} \tag{68}$$

4. Interpretation

- $\alpha \uparrow$: more weight on energy reduction
- $\alpha \downarrow$: more weight on battery degradation reduction
- `obj_w_trac` and `obj_w_cool`: split the energy term between traction and cooling

Nonlinear Constraints: get_MPCsolver.m

1. Power and thermal constraints

$$P_{\text{req},k} \leq P_{\max} \tag{69}$$

$$Q_{\text{joule},k} \leq Q_{\max} \tag{70}$$

$$P_{\text{cool},k} \leq P_{\text{cool},\max} \tag{71}$$

$$T_{\text{batt},\min} \leq T_{\text{batt},k} \leq T_{\text{batt},\max} \tag{72}$$

2. Terminal distance constraint

$$|d_{\text{end}} - d_{\text{ref}}| \leq \varepsilon_{\text{dist}} \tag{73}$$

Gap-Based Bounds: get_bounds.m

1. Speed bounds from front and rear gaps

$$\bar{v}_{k+1} = v_{F,k} + \frac{d_{F,k} - d_{\mathrm{tol}}}{\Delta t} \tag{74}$$

$$\underline{v}_{k+1} = v_{R,k} + \frac{d_{\mathrm{tol}} - d_{R,k}}{\Delta t} \tag{75}$$

2. Acceleration bounds

$$\bar{a}_k = \frac{\bar{v}_{k+1} - v_k}{\Delta t} \tag{76}$$

$$\underline{a}_k = \frac{\underline{v}_{k+1} - v_k}{\Delta t} \tag{77}$$

3. Clipping and relaxation

- no front vehicle: $\bar{a}_k \leftarrow a_{\mathrm{tol}}$
- no rear vehicle: $\underline{a}_k \leftarrow -a_{\mathrm{tol}}$
- all bounds are clipped to $[-a_{\mathrm{tol}}, a_{\mathrm{tol}}]$

Linear Inequality System: ABineq

1. Decision variable

$$\boldsymbol{x} = \begin{bmatrix} v_2 & \cdots & v_{N_p+1} \end{bmatrix}^\top \tag{78}$$

2. Acceleration constraints

$$\underline{a}_k \leq \frac{v_{k+1} - v_k}{\Delta t_k} \leq \bar{a}_k \tag{79}$$

3. Speed box constraints

$$\underline{v}_k \leq v_{k+1} \leq \bar{v}_k \tag{80}$$

Linear Inequality Matrix Form: ABineq

4. Matrix form

$$A_{\text{ineq}}\boldsymbol{x} \leq \boldsymbol{b}_{\text{ineq}} \tag{81}$$

- rows 1–N_p: upper acceleration constraints
- rows $N_p + 1$–$2N_p$: lower acceleration constraints
- rows $2N_p + 1$–$3N_p$: upper speed constraints
- rows $3N_p + 1$–$4N_p$: lower speed constraints

Thermal Parameters: get_parameters.m

Temperature settings

$$T_{\text{amb}} = 25^\circ\text{C}, \qquad T_{\text{init}} = 32^\circ\text{C}, \qquad T_{\text{ref}} = 20^\circ\text{C} \tag{82}$$

$$T_{\text{batt,min}} = 15^\circ\text{C}, \qquad T_{\text{batt,max}} = 45^\circ\text{C} \tag{83}$$

Cooling settings

$$P_{\text{cool,max}} = 4000\ \text{W} \tag{84}$$

$$\eta_{\text{cool}} = 1.5, \qquad hA = 0 \tag{85}$$

Thermal capacity

$$C_{\text{th}} = m_{\text{bat}} c_p \tag{86}$$

Logging Structure: get_HistoryInit.m / get_HistoryLog.m

Prediction logs

- `history.pred.pre`: baseline prediction window
- `history.pred.opt`: optimized prediction window
- stored fields: speed, acceleration, power, current, SOC, temperature, aging, and bounds

Replay logs

- `history.app`: applied optimized trajectory
- `history.base`: replay-baseline trajectory
- meta fields: previous speed, traveled distance, initial position, and battery state

Edit Order When EV Outputs Change

Recommended update order

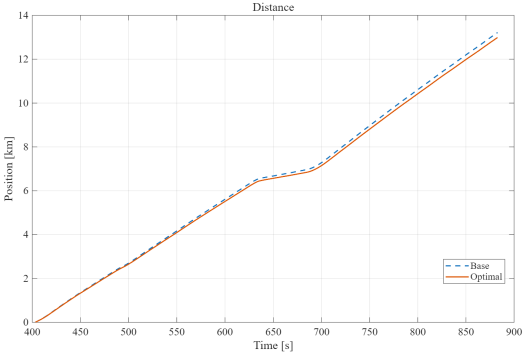
- `get_evmodel.m`: add or modify `EVout.xxx`
- `get_pred_hist.m`: add the prediction-side field
- `get_appd_hist.m`: add the replay-side field
- `get_plot.m`: update figures only if visualization is needed

Current output groups

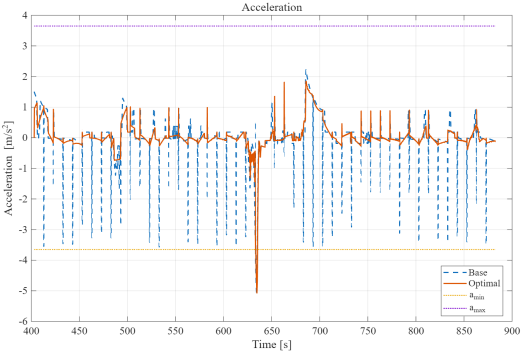
- power: `P_req`, `P_trac`, `P_cool`
- battery: `I_batt`, `Q_Ah`, `SOC`, `Rin`
- thermal: `T_batt`, `Q_batt`, `Q_env`, `Q_cool`
- aging: `Ah_ir`, `Q_loss`, `Ah_eff`, `AgeCost`

Vehicle Motion Results

Distance history

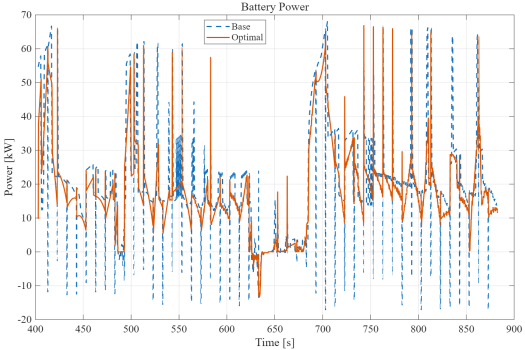


Acceleration history

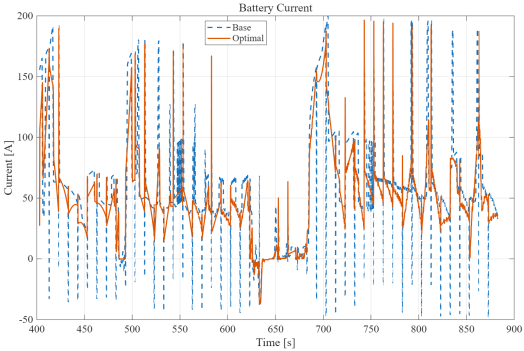


Electrical Load Results

Power demand

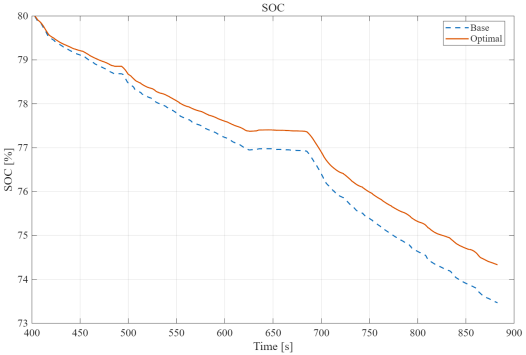


Battery current

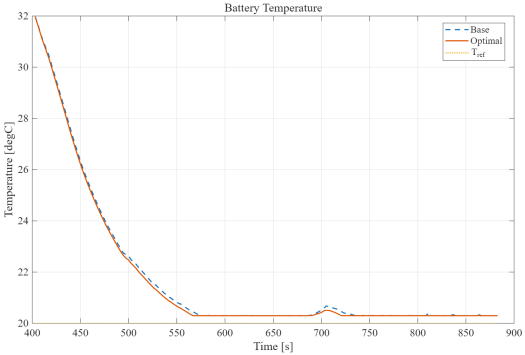


Battery State Results

State of charge

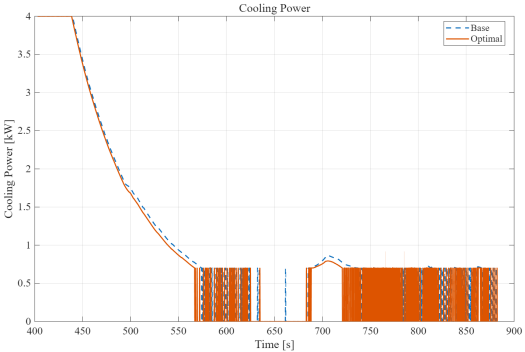


Battery temperature

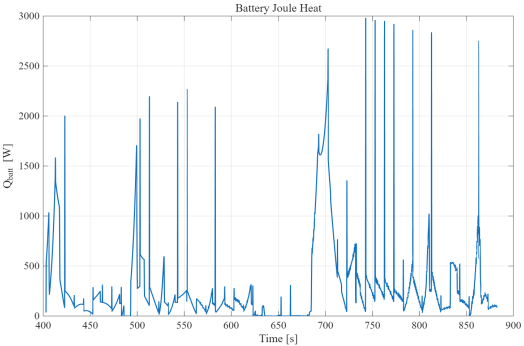


Thermal Management Results

Cooling power

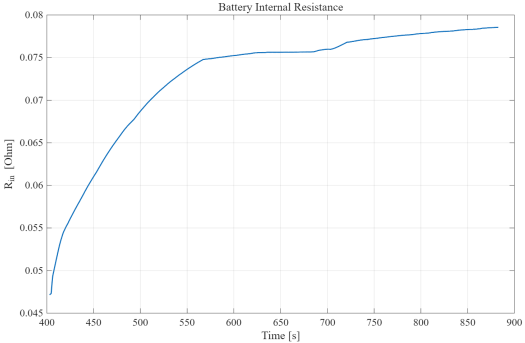


Battery heat generation

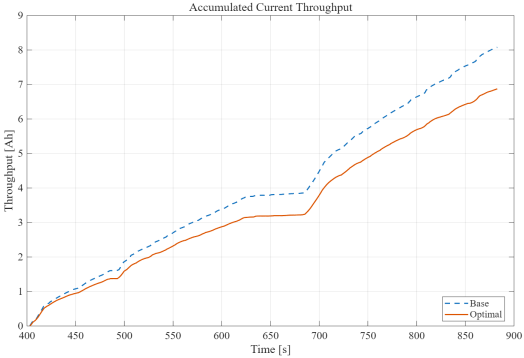


Aging State Results

Internal resistance

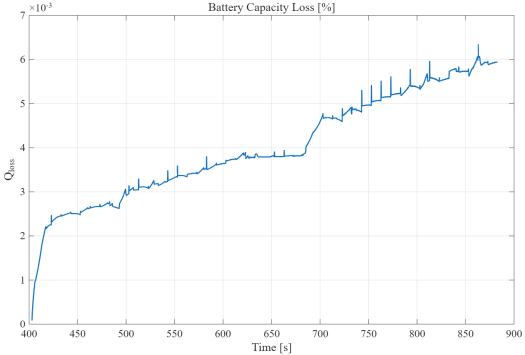


Accumulated Ah throughput

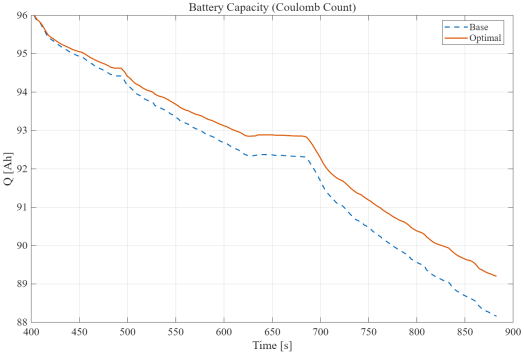


Degradation Results

Capacity loss



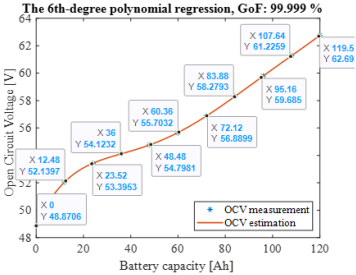
Remaining capacity



SOC에서 OCV 구하기

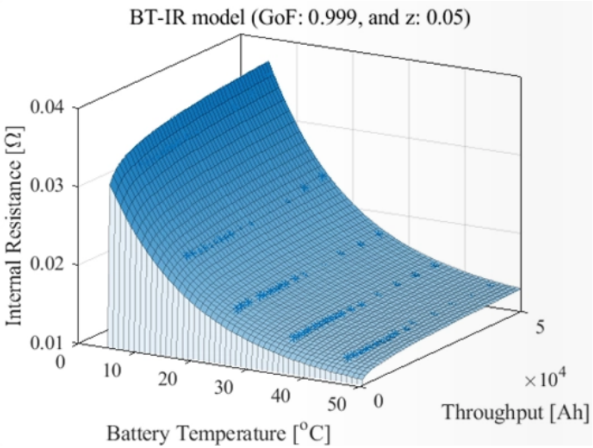
```
x_set1 ; % 배터리 용량 [Ah]
Load T1_OCV_AH_Data_module2_mu.mat % 변수 mu
Load T1_OCV_AH_Data_module2_sys.mat % 변수 p

x_norm = (x_set1 - mu(1))/mu(2);
est_OCV =
p(1)*(x_norm.^6)+p(2)*(x_norm.^5)+p(3)*(x_norm.^4)+
p(4)*(x_norm.^3)+p(5)*(x_norm.^2)+p(6)*(x_norm)+p(7);
```



Internal Resistance Model

3. Fit_IR: 내부저항 모형 (입력 파라미터: 누적 사용량 [Ah], 배터리 온도)



Q/A